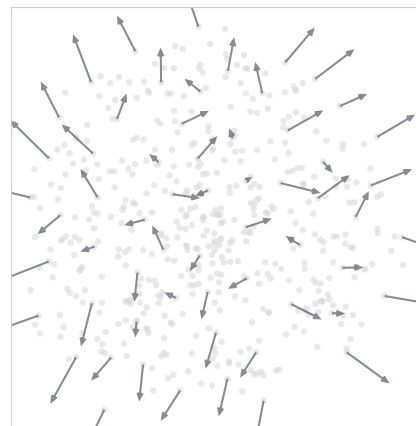
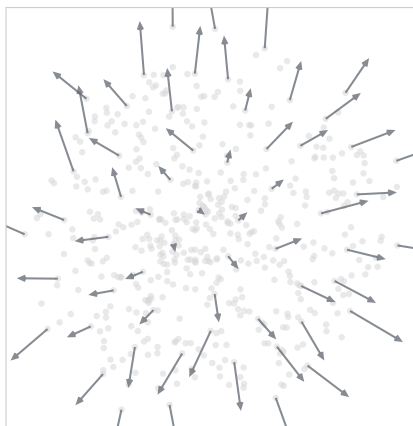
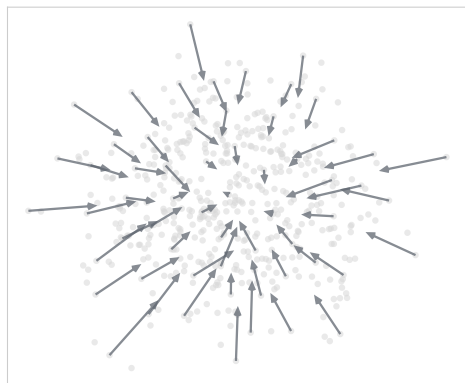


**Pre-grokking  
(1500 → 2000)**

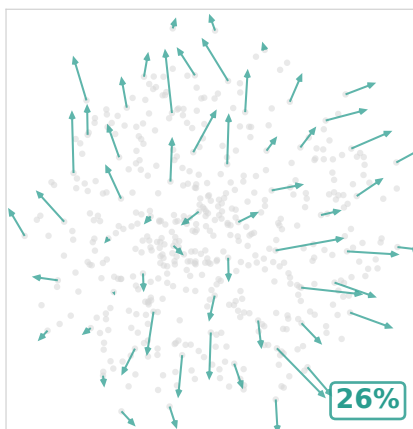
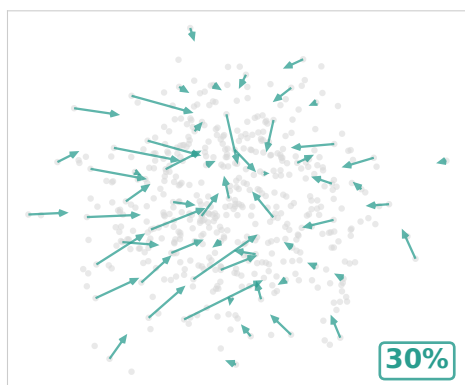
**Transition  
(2500 → 3000)**

**Post-grokking  
(4000 → 4500)**

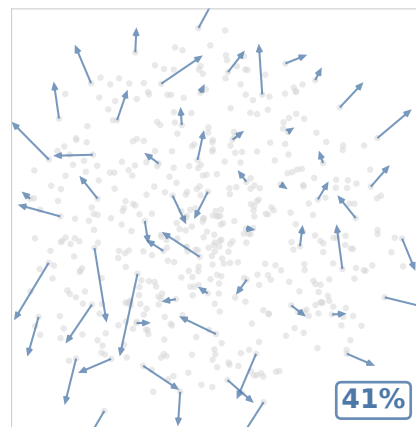
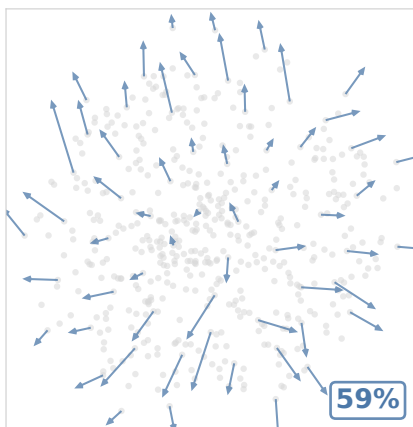
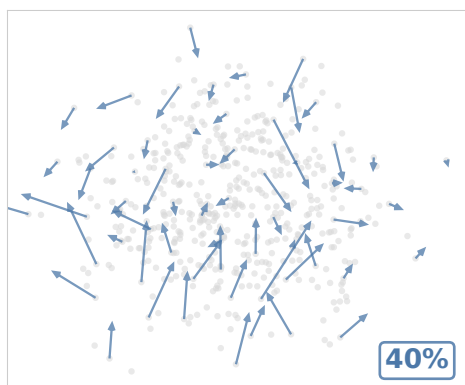
**Total**



**Exact (gradient)**



**Coexact (curl)**



**Harmonic**

